Note of Fast Accurate Eigenvalue Computations Using The Cholesky Factorization

Theorem 2.3:
Let $G = BD \in \mathbb{R}^{n\times n}$ be nonsingular and let $\hat{G} = (B + \Delta B)D$ where $\|\Delta B\| \le \sigma_n(B)$.

$$\left|\frac{\sigma_i(\hat{G}) - \sigma_i(G)}{\sigma_i(G)}\right| \le \frac{\|\Delta B\|}{\sigma_n(B)}, \quad i = 1, \ldots, n.$$

Co

$$G = L\tilde{D}^{-1}\tilde{D}$$
$$\tilde{G} = (L\tilde{D}^{-1} - \Delta\tilde{D}^{-1})\tilde{D} = L_{(1)}\tilde{D}^{-1}\tilde{D}$$
$$\left\|L\tilde{D}^{-1}\right\| \le \frac{\left\|\Delta\tilde{D}^{-1}\right\|}{\sigma_n(L\tilde{D}^{-1})}$$

1 Derivative of eigenvalues and eigenvectors

Derivative of an eigenvalue $\lambda(X)$ and an eigenvector $u(X)$ of matrix X at X_0.

$\cdot^+$ means Moore-Penrose inverse.

TODO:

1. Relation of these two expressions?
2. SVD when has non-simple roots?
3. Derivative of normalization operation?

1.1 Real symmatric case

$X_0 \in \mathbb{R}^{n\times n}$, $X_0 = X_0^T$:

$$X_0 u_0 = \lambda_0 u_0$$

and λ_0 be a simple eigenvalue, then

$$Xu = \lambda u, \quad u^T u = 1, \quad X \in N(X_0)$$

$$\mathrm{d}\lambda = u_0^T(\mathrm{d}X)u_0$$
$$\mathrm{d}u = (\lambda_0 I_n - X_0)^+(\mathrm{d}X)u_0$$

1.2 General case

1.2.1 Normalization $u_0^* u = 1$

See [2]

$Z_0 \in \mathbb{C}^{n\times n}$:

$$Z_0 u_0 = \lambda_0 u_0$$

and λ_0 be simple eigenvalue, then

$$Zu = \lambda u, \quad u_0^* u = 1, \quad Z \in N(Z_0)$$

$$\mathrm{d}\lambda = \frac{v_0^*(\mathrm{d}Z)u_0}{v_0^* u_0}$$

$$\mathrm{d}u = (\lambda_0 I - Z_0)^+ \left(I - \frac{u_0 v_0^*}{v_0^* u_0}\right)(\mathrm{d}Z)u_0 \tag{1}$$

where $Z_0^* v_0 = \bar{\lambda}_0 v_0$, i.e. v_0^* is the left eigenvector of Z_0 corresponding to λ_0.

1.2.2 Normalization diagC=0

See [1]

Assume: $A \in \mathbb{C}^{n\times n}$?, $\lambda_i \neq \lambda_j \ \forall i \neq j$. $AU = U\Lambda$.

$$\mathrm{d}\Lambda = I \circ \left(U^{-1}\mathrm{d}A\, U\right)$$

Normalization: $\mathrm{diag}\left(U^{-1}\mathrm{d}U\right) = 0$:

$$\mathrm{d}U = U\left(F \circ \left(U^{-1}\mathrm{d}A\, U\right)\right) \tag{2}$$

where $F_{ij} = (\lambda_j - \lambda_i)^{-1}$ for $i \neq j$, and zero otherwise.

In general, Eq.(1) is different from Eq.(2). Eq.(2) seems has problem with complex conjugate roots. The authur only verified real eigenvalue cases. Also it needs extra normalization step to get accurate result.

2 Derivative of SVD

See [1].

Assume $A \in \mathbb{R}^{m\times n}$,

$$A = USV^T,$$

then

$$\mathrm{d}S = I \circ \left(U^T \mathrm{d}A\, V\right).$$

Assume $m \leq n$.

Defined $\mathrm{d}C = U^{-1}\mathrm{d}U$, $\mathrm{d}D = V^{-1}\mathrm{d}V$ (note $\mathrm{d}C$ and $\mathrm{d}D$ are anti-symmetric), and partition

$$S = \begin{bmatrix} S_1 & 0 \end{bmatrix}, \quad \mathrm{d}S = \begin{bmatrix} \mathrm{d}S_1 & 0 \end{bmatrix}, \quad \mathrm{d}D = \begin{bmatrix} \mathrm{d}D_1 & -\mathrm{d}D_2 \\ \mathrm{d}D_2^T & \mathrm{d}D_3 \end{bmatrix}$$

$$U^T \mathrm{d}A\, V = \begin{bmatrix} \mathrm{d}P_1 & \mathrm{d}P_2 \end{bmatrix}$$

then

$$\mathrm{d}D_2 = S_1^{-1}\mathrm{d}P_2$$
$$\mathrm{d}C = F \circ \left(\mathrm{d}P_1 S_1 + S_1 \mathrm{d}P_1^T\right)$$
$$\mathrm{d}D_1 = F \circ \left(S_1 \mathrm{d}P_1 + \mathrm{d}P_1^T S_1\right)$$

where $F_{ij} = \left(s_j^2 - s_i^2\right)^{-1}$ for $i \neq j$ and zero otherwise.

$$\mathrm{d}U = U\left(F \circ \left(\mathrm{d}P_1\, S_1 + S_1 \mathrm{d}P_1^T\right)\right)$$

$$\begin{bmatrix}\mathrm{d}V_1 \\ \mathrm{d}V_2\end{bmatrix} = V \begin{bmatrix}\mathrm{d}D_1 \\ \mathrm{d}D_2^T\end{bmatrix}$$

3 Derivative of Cholesky decomposition

Suppose $\Sigma = \Sigma^T \in \mathbb{R}^{n\times n}$,

$$LL^T = \Sigma$$

with perturbation $\mathrm{d}\Sigma = \mathrm{d}\Sigma^T$, then

$$\mathrm{d}L = L\,\Phi(L^{-1}\mathrm{d}\Sigma\left(L^{-1}\right)^T)$$

where the Φ operator extract the lower-triangular part of a matrix and halves its diagonal:

$$\Phi_{ij}(A) = \begin{cases} A_{ij} & i > j \\ \frac{1}{2}A_{ii} & i = j \\ 0 & i < j \end{cases}$$

References

[1] Mike Giles. An extended collection of matrix derivative results for forward and reverse mode algorithmic differentiation. page 23.

[2] Jan R. Magnus. On Differentiating Eigenvalues and Eigenvectors. *Econometric Theory*, 1(2):179–191, August 1985.